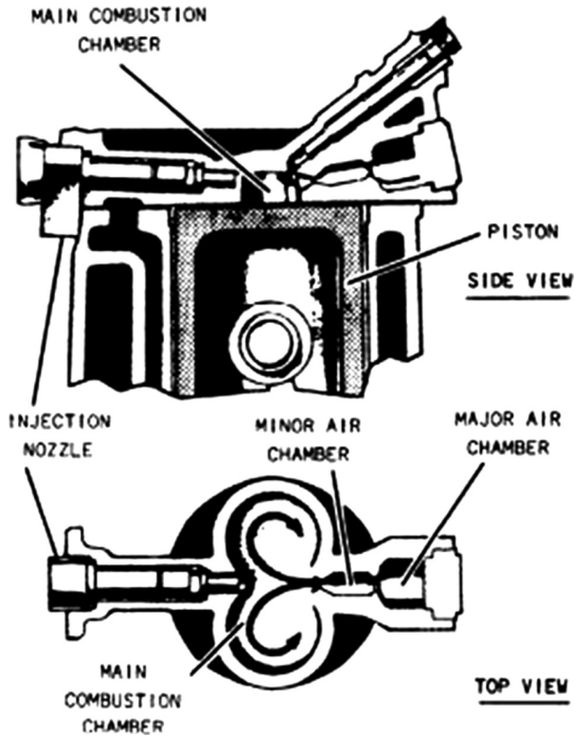


Figure 6.11.1(e) Air Cell and Energy Cell

are connected together by a narrow passage. Due to this arrangement the pressure of the inlet air increases in the air cell chamber and the pressurized air enters into the main chamber in which the fuel is injected and combustion is achieved. Figure 6.11.1(e) shows the Air Cell and Energy Cell.

Energy cell

It is also known as Lenovo combustion chamber. It combines the Air cell and Pre combustion chamber modes. In this type of chamber the energy cell is connected via the narrow passage with main chamber. When the engine runs, the pressure of the air in the energy cell is increased due to the narrow path and fuel is also injected in the similar way as pre chamber. The combustion at the energy cell is continued in the main chamber where the main combustion takes place. By this way the energy is utilized by injecting fuel in the energy cell. Due to piston movement from TDC to BDC and depending on pressure difference between the chambers the mixture is combusted completely.

6.12 EXHAUST SYSTEM

In an engine the products of combustion (pollutants) formed during the combustion process must be taken out of the engine. This system is called as the exhaust system. This system includes exhaust port, exhaust manifold, exhaust pipe, muffler and a catalytic converter.

6.12.1 Exhaust manifold

Exhaust manifold is a part of the exhaust system that is used for removing unwanted gases such as carbon monoxide, carbon dioxide, smoke, unburned hydrocarbon etc., which are formed during the combustion process. Through the exhaust port of the engine the exhaust products are passed to the exhaust manifold. The unwanted gases coming out from the all cylinders in the engine are collected in the exhaust manifold and sent to the exhaust pipe called as tail pipe. This is the main purpose of the exhaust manifold present in the engine. Exhaust manifold is generally made up of cast iron. Figure 6.12.1 shows the Exhaust Manifold.

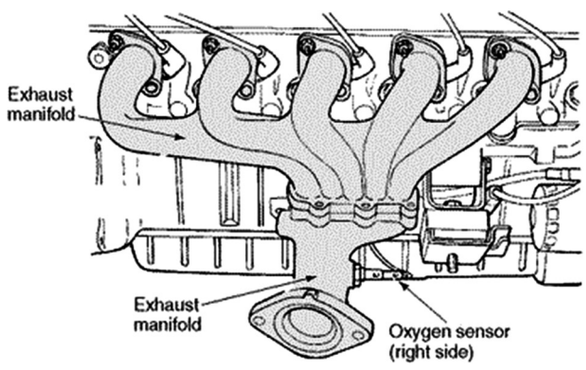


Figure 6.12.1 Exhaust Manifold

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